

$$X_n(x) = \sin(n\pi x/L), \qquad \lambda_n = (n\pi/L)^2,$$

$$T_n(t) = \exp(-\kappa\lambda_nt),$$

$$v(\xi,\tau) = e^{-\pi^2\tau}\sin(\pi\xi) + 0.35e^{-9\pi^2\tau}\sin(3\pi\xi),$$

$$E(t) = \tfrac{1}{2} \int_0^L u^2 \, dx,$$

$$\frac{dE}{dt} = -\kappa \int_0^L (u_x)^2 \, dx \leq 0.$$